flown by the Earth a hundred million years ago, in the age of the dinosaurs when there were no humans and no technology, you would still have seen oxygen and ozone, the chlorophyll pigment, and far too much methane. At present, though, your instruments are finding signs not just of life, but of high technology—something that couldn't possibly have been detected even a hundred years ago:

You are detecting a particular kind of radio wave emanating from Earth. Radio waves don't necessarily signify life and intelligence. Many natural processes generate them. You've already found radio emissions from other, apparently uninhabited worlds—generated by electrons trapped in the strong magnetic fields of planets, by chaotic motions at the shock front that separates these magnetic fields from the interplanetary magnetic field, and by lightning. (Radio "whistlers" usually sweep from high notes to low, and then begin again.) Some of these radio emissions are continuous; some come in repetitive bursts; some last a few minutes and then disappear.

But this is different: A portion of the radio transmission from Earth is at just the frequencies where radio waves begin to leak out of the planet's ionosphere, the electrically charged region above the stratosphere that reflects and absorbs radio waves. There is a constant central frequency for each transmission, added to which is a modulated signal (a complex sequence of ons and offs). No electrons in magnetic fields, no shock waves, no lightning discharges can generate something like this. Intelligent life seems to be the only possible explanation. Your conclusion that the radio transmission is due to technology on Earth holds no matter what the ons and offs mean: You don't have to decode the message to be sure it *is* a message. (This signal is really, let us suppose, communications from the U.S. Navy to its distant nuclear-armed submarines.)

So, as an alien explorer, you would know that at least one species on Earth has achieved radio technology. Which one is it? The beings that make methane? Those that generate oxygen? The ones whose pigment colors the landscape green? Or somebody else, somebody more subtle, someone not otherwise detectable to a spacecraft plummeting by? To search for this technological species, you might